SAMRAT ASHOK TECHNOLOGICAL INSTITUTE, VIDISHA SET A

MID SEM EXAM 2ND, BE I[ST] SEM. APP. PHYSICS,PYB101,IT

Note: - Attempt one part from each question.

		MARK	BL	CO
Q.1	WHAT IS FERMI ENERGY LEVEL	2	1	3
OR	WRITE TWO MAIN PROPERTIES OF PN JUNCTION.	2	1	3
Q.2	WRITE DOWN THE THREE IMPORTANT APPLICATIONS OF SUPERCONDUCTORS.	3	2	4
OR	EXPLAIN THE MEISSNER EFFECT OF SUPERCONDUCTOR.	3	2	4
Q.3	OBTAIN THE RELATION BETWEEN E, D AND P VECTOR.	5	3	5
OR	[illegible]	5	3	5

SAMRAT ASHOK TECHNOLOGICAL INSTITUTE, VIDISHA SET B

MID SEM EXAM 1ST, BE 2ND SEM. APP. PHYSICS,PYB101 IT

Note: - Attempt one part from each question.

		MARK	BL	CO
Q.1	DEFINE DE BROGLIE MATTER WAVE.	2	1	3
OR	DEFINE PHOTON WITH ITS UNIT.	2	1	3
Q.2	EXPLAIN THE STIMULATED EMISSION OF RADIATION.	3	2	4
OR	WHAT IS NUMERICAL APERATURE.WRITE ITS FORMULA.	3	2	4
Q.3	DIFFERENTIATE BETWEEN THE DIRECT AND INDIRECT BAND GAP OF SEMICONDUCTOR.	5	3	5
OR	EXPLAIN THE RADIATIVE AND NON RADIATIVE RECOMBINATION MECHANISM.	5	3	5